Nephele - Remote Healthcare Services, Ultrasound Medical Device Dematerialization

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***Abstract*—The NEPHELE project, funded by the Horizon Europe programme under the topic "Future European platforms for the Edge: Meta Operating Systems," aims to revolutionize the orchestration of Hyper-Distributed Applications (HDAs) across the compute continuum from Cloud-to-Edge-to-IoT (Internet of Things). By addressing existing openness and interoperability barriers, NEPHELE introduces automation and decentralized intelligence mechanisms powered by 5G and distributed Artificial Intelligence (AI) technologies. The project focuses on two core innovations: an IoT and edge computing SoftWare (SW) stack that supports virtualization and interoperability in a device-independent manner, and a synergetic meta-orchestration framework for coordinating cloud and edge computing platforms. These innovations will be demonstrated and validated across various vertical industries, including disaster management, logistics, energy management, and remote healthcare. In this context, our work demonstrates the process of virtualizing the core functionalities of conventional ultrasound medical systems, effectively abstracting away physical constraints and facilitating remote Ultrasound exams via edge/cloud services. This paper presents a comprehensive analysis of Remote Ultrasound Medical Services, focusing on its IoT components and services, highlighting the positive impact on sustainability, cost reduction and increased flexibility, despite slight reductions in performance metrics.**

Keywords—Ultrasound Medical Device, Dematerialization, Edge, Cloud, IoT, Hyper-Distributed Application

I. Introduction

In the Ultrasound (US) medical area, the standalone and featured all-in-one solution paradigm has been prevalent since the early history of ultrasound medical devices. The evolution of ultrasound equipment has been marked by advancements in both hardware and software. Over the decades, two key aspects have emerged:

- **Miniaturization and Power Enhancement**: there has been a continuous trend towards the miniaturization of hardware components, coupled with significant increases in their power.
- **Shift from Hardware to Software Processing**: the processing unit has transitioned from being hardware-based to software-based, leveraging powerful Central Processing Units (CPUs) and Graphics Processing Units (GPUs).

Despite these advancements, the paradigm remained largely unchanged, maintaining a standalone solution approach. Forever, in recent years, a shift towards network connectivity has begun to emerge. This new approach offers several advantages [1], including:

- **Data and Feature Sharing**: the ability to share data and features through centralized services accessible from different parts of the world.
- **Cost Reduction**: lowering the costs of onboard hardware components (such as CPUs and GPUs for heavy processing) by moving the processing unit to a dedicated service.

The Remote Healthcare Services (RHSs) in Nephele project focuses on the dematerialization of ultrasound scanners. The goal is to transform ultrasound acquisition HardWare (HW) and medical imaging viewers into smart, wireless-connected devices that can be plugged and played through a cloud-edge medical imaging application. This involves decomposing and virtualizing ultrasound medical imaging systems into a cloud-edge continuum, eliminating barriers due to hardware capabilities and localization of current physical systems. The vision aims to significantly lower acquisition costs, promote local health even in rural areas, and introduce novel functionalities for augmenting medical analysis enabled by cloud-edge powered Artificial Intelligence (AI) [2].

The remainder of this paper is organized as follows: **Section II** provides an overview of the NEPHELE platform, including its use cases and key components. **Section III** discusses the dematerialization of ultrasound medical devices, detailing the actors in Remote Ultrasound Medical Services and the Application Graph within NEPHELE. **Section IV** evaluates performance, examining latency, bandwidth consumption, compression efficiency, and Quality of Service (QoS) enhancements. Finally, **Section V** concludes, summarizing key achievements and future challenges.

II. Nephele

The NEPHELE project introduces an Internet of Things (IoT) and edge/cloud computing software stack that leverages the virtualization of IoT devices at the edge of the infrastructure, utilizing **Kubernetes for container orchestration and management**.

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A. Overview of Use Cases

The NEPHELE platform supports various use cases, including a wide range of applications across different areas:

- **Emergency/disaster recovery**: this use case involves the deployment of ground robots and drones to map affected areas, detect victims, assess their injuries, predict risks, and deploy necessary devices and liquid samplers.

  To enhance the efficiency and effectiveness of emergency response operations by leveraging advanced technologies for Real-Time (RT) mapping, victim detection, and risk assessment.

- **AI-assisted logistics operations in the Port of Koper**: this use case focuses on optimizing logistics operations in the Port of Koper using AI technologies.

  To improve the efficiency of logistics operations, reduce operational costs, and enhance the overall productivity of the port.

- **Energy management in smart buildings/cities**: this use case involves intelligent monitoring and remote energy management in a Smart Building testbed scenario.

  To optimize energy consumption, reduce energy costs, and enhance the sustainability of smart buildings and cities.

- **RHSs**: this use case focuses on providing Ultrasound Medical Device Dematerialization through NEPHELE platform.

  To enhance healthcare delivery by enabling remote support, and treatment of patients, improving accessibility, reducing costs, and increasing flexibility.

These use cases demonstrate the versatility and potential of the NEPHELE platform in various domains. Each use case leverages the platform's advanced technologies to address specific challenges and achieve significant improvements in efficiency, cost-effectiveness, and sustainability, addressing different real-world scenarios.

B. Components

Below are the key components of the NEPHELE platform. These components are integral to the platform's architecture, ensuring robust performance, scalability, and seamless integration. Each component plays a vital role in delivering the platform's functionalities [4]:

- **Virtual Objects (VO)**: digital representations of physical devices or sensors. Each VO is responsible for capturing specific data and sending it to the central system for processing.
- **Composite Virtual Objects (CVO)**: higher-level abstractions that combine multiple VOs to provide more complex functionalities. For example, a CVO might aggregate data from various monitoring devices to provide a comprehensive view of a specific scenario. CVOs enable the system to perform more sophisticated analyses and generate insights that are not possible with individual VOs alone.
- **Virtual Object Stack (VOStack)**: multi-layered lightweight software stack designed for the virtualization of IoT devices and functions at the edge. It aims to tackle IoT technologies' convergence and interoperability aspects, support distributed data management and analysis, exchange of information, autonomic networking, and ad-hoc group management functionalities.
- **Application Components**: software modules and services that process the data collected by VOs and CVOs. They implement the business logic and provide the necessary functionalities for specific scenarios. Key application components might include data processing algorithms, machine learning models for prediction, user interfaces, and communication modules for real-time data transmission.

III. Ultrasound Medical Device Dematerialization

RHSs aim to enhance healthcare delivery by leveraging IoT components, virtualization, and orchestration frameworks to enable remote support and treatment of patients.

A. Main Actors

Below are the actors involved in a remote US Exam. These actors play crucial roles in ensuring the successful execution and accuracy of the examination. Each actor's responsibilities and interactions are vital for the seamless operation of the remote ultrasound process [3]:

- **Patients**: Individuals receiving healthcare services remotely (probe data in Fig. 1).
- **Local Healthcare Providers:** Doctors and operators near patients (probe data, minimal HW/SW and Web Application Dashboard in Fig. 1).
- **Remote Healthcare Providers**: Doctors and other medical professionals remotely deliver care (Web Application Viewer in Fig. 1).
- **IoT Device**: Ultrasound Medical device used for monitoring patients' health (minimal HW/SW in Fig. 1).
- **NEPHELE Platform**: The underlying infrastructure that supports data collection, processing, and communication between local and remote healthcare providers (VO Stack, VO, Video Streaming Service and Message Broker in Fig. 1).

Fig. 1 shows the interaction among actors.

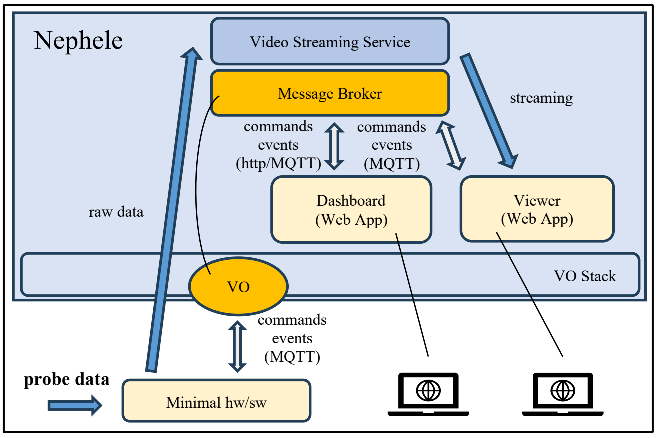

Figure 1. Actors Interaction during a Remote Ultrasound Exam

The scenario shows a Real-Time US exam performed by an operator with remote support provided by a doctor. The connection between the US device and NEPHELE is based on a Virtual Object. The operator can connect US equipment to the NEPHELE infrastructure to make the device available for remote communication. The US connection is facilitated by a Device Portal that provides all information related to the endpoints and the service used.

Once the US equipment is connected, remote operators can see the device available for remote usage. Remote operators can handle parameters during real-time exams using the Web Application Dashboard and watch the exam using the Web Application Viewer. Parameters such as focus, depth, density, and gain can be adjusted to improve the quality of the image for better capture of issues or diseases. Communication among components is provided by a dedicated Message Broker.

Remote operators can monitor their actions through real-time video streaming from US equipment. The video streaming broadcasting is provided by a dedicated service (Video Streaming Service).

B. *Application Graph*

In this section, a detailed explanation presents components involved in the Application Graph (AG) for RHSs (Fig. 2). The AG is a critical element in the design and implementation of complex systems based on Nephele Platform, enabling the visualization and management of various interconnected components. Each component within the AG plays a specific role, contributing to the overall functionality and performance of the system. Understanding these components is essential for optimizing system efficiency and ensuring seamless integration.

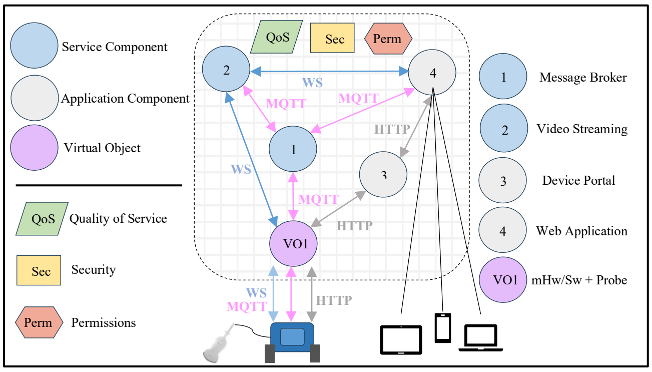

Figure 2. Ultrasound Medical Device Dematerialization Application Graph

Here is the explanation of components involved in AG:

- **VO: Minimal HW/SW (IoT)**: the US equipment and the probe, the hardware part that produces the analog signal, is connected to NEPHELE thanks to the VO abstraction layer. The IoT can send data related to telemetry and streaming video through VO. In addition, communication to the Dashboard Web Application passes through VO.
- **Application Component: Web Application Viewer**: it shows the video exam streaming in real-time and offline (when the exam comes from the archive) modes.
- **Application Component: Web Application Dashboard**: it provides the capability to control US physical devices remotely.
- **Application Component: Video Streaming Service**: this service works as a bridge between US equipment and all clients connected to share the Real-Time exam.
- **Application Component: Message Broker**: it is a service for communication between a client and US equipment. The client can manage the device remotely. At the same time the operator in front of the device can manage the device itself. Message Broker synchronizes the actors sending updated messages.
- **Application Component: Device Portal:** the portal provides the entry point for the US equipment for registration and connection. At the same time, it provides. the entry point also for clients, giving them all information about Message Broker type and communication endpoints.

Secure communication among all actors is ensured by default. This is achieved through robust encryption protocols and stringent access controls. Each data exchange is authenticated and encrypted, preventing unauthorized access and ensuring data integrity. Our approach guarantees that all interactions within the IoT ecosystem are protected against potential threats. This secure-by-default design is fundamental to maintaining the confidentiality and reliability of the system [5].

IV. Performance Evaluation

While traditional standalone ultrasound equipment, operating without network dependency, offers superior performance and real-time responsiveness, cloud-based solutions introduce new possibilities for remote ultrasound exam performing and resource optimization.

To assess the feasibility and efficiency of the proposed cloud-based ultrasound medical solution, we conducted controlled experiments within a simulated healthcare environment. The testbed consisted of a virtualized cloud server running the ultrasound processing system, an edge computing node for real-time data relay, and a client interface representing the remote operator. The evaluation considered variations in **network latency** and **bandwidth consumption** on remote ultrasound exam usability.

A. *Metrics and Key Performance Indicators (KPIs)*

The following key performance metrics were analyzed to validate system performance (as shown in Table 1):

- **Latency**: Defines the time delay between probe movement on a patient's skin and image rendering on the remote operator's display.

  KPI: Maintain latency below 100ms for acceptable performance, with an optimal threshold of 40ms to avoid perceptible delays.

- **Bandwidth Consumption**: Evaluates the data rate required for high-quality image transmission.

  KPI: A 1200×760-pixel ultrasound frame required 90 MB/sec of bandwidth for accurate rendering.

- **Compression Efficiency**: Determines the potential reduction in bandwidth usage through image lossless compression.

  Observation: Since a significant portion of ultrasound images is black, compression algorithms reduce bandwidth demand without degrading image quality.

- **Quality of Service (QoS) and Network Slicing**: Investigates techniques to enhance stability and reliability.

  Observation: Dedicated network slices for ultrasound imaging improve data transmission consistency, ensuring minimal interference from other services.

TABLE I. KPIS RANGES

Metric	***Optimal Value***	***Acceptable Range***	***Critical Threshold***
Latency (ms)	<= 40ms	40ms – 100ms	> 100ms (not usable)
Bandwidth (MB/sec)	90 MB/sec	60MB/sec – 90 MB/sec	< 60MB/sec (image degradation)
Compression Efficiency	35% data reduction	25% - 40% reduction	< 25% reduction (minimal impact)
QoS Optimized Slicing	Stable performance	Minor variations	High network traffic interference

B. *Discussion of Findings*

The evaluation confirms that latency and bandwidth availability are critical factors in enabling a functional cloud-based ultrasound system. While maintaining latency below 100ms is feasible for remote diagnostic operations, achieving the optimal threshold of 40ms requires significant advancements in network infrastructure.

Historically, legacy networks struggled to support the real-time demands of high-resolution medical imaging. However, the evolution from **5G to emerging 6G technologies** offers transformative potential. The introduction of **ultra-reliable low-latency communications (URLLC)** in **5G networks** allows for precise data transmission with latencies approaching the required 40ms threshold. Additionally, network slicing, a key feature of modern networking architectures, enables dedicated bandwidth allocation for medical imaging, ensuring minimal interference from non-critical services.

Beyond latency, bandwidth remains a major challenge. A 1200×760-pixel ultrasound frame requires 90 MB/sec, posing significant demands on network capacity. The integration of **adaptive compression algorithms** can mitigate this issue by reducing unnecessary data transmission, particularly by exploiting the large black areas inherent in ultrasound images. Furthermore, **edge computing infrastructures**—which process medical imaging closer to the source rather than relying solely on centralized cloud servers—can further reduce transmission delays and optimize performance.

These findings highlight that while **cloud-based ultrasound imaging is technologically feasible**, sustained improvements in **network infrastructure, QoS optimization, and edge-cloud synergies** will be essential to ensure real-time, high-quality diagnostic capabilities. Future advancements in **5G, AI-driven network management, and next-generation 6G architectures** will play a pivotal role in shaping the next era of remote medical services.

V. CONCLUSION

The dematerialization of ultrasound scanners offers significant benefits from various perspectives. Improved accessibility ensures that remote healthcare services make medical care available to patients in underserved areas. Cost reduction is achieved by minimizing the need for physical infrastructure and in-person visits, leading to substantial savings. Enhanced patient monitoring allows for the early detection of health issues, thereby improving patient outcomes. Increased flexibility enables healthcare providers to deliver care from any location, enhancing their responsiveness to patients' needs.

However, the dematerialization of ultrasound scanners also presents challenges that must be addressed. Ensuring data privacy and security is paramount to protect patients' health information. Interoperability is crucial for integrating various services and ensuring seamless operation with the NEPHELE platform. Lastly, the reliability of the NEPHELE platform must be maintained to handle the continuous influx of health data without downtime. In addition, further research should focus on enhancing compression techniques, improving real-time processing mechanisms, and refining network slicing strategies to ensure sustained performance and reliability across diverse clinical environments.

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